KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

WEB TECHNOLOGY – ASSIGNMENT II

Course Code: U21CS501

Project Title

Student Admission Portal with Profile & Document Management

Submitted by

Mithuna Kamalanathan

Roll No: 24CS117

B.E CSE

Academic Year

2026–2027

QUESTION 1 – Student Registration Form (8 Marks)

Aim

To create a student registration form using HTML, style it with CSS and validate the entered details with JavaScript.

Implementation

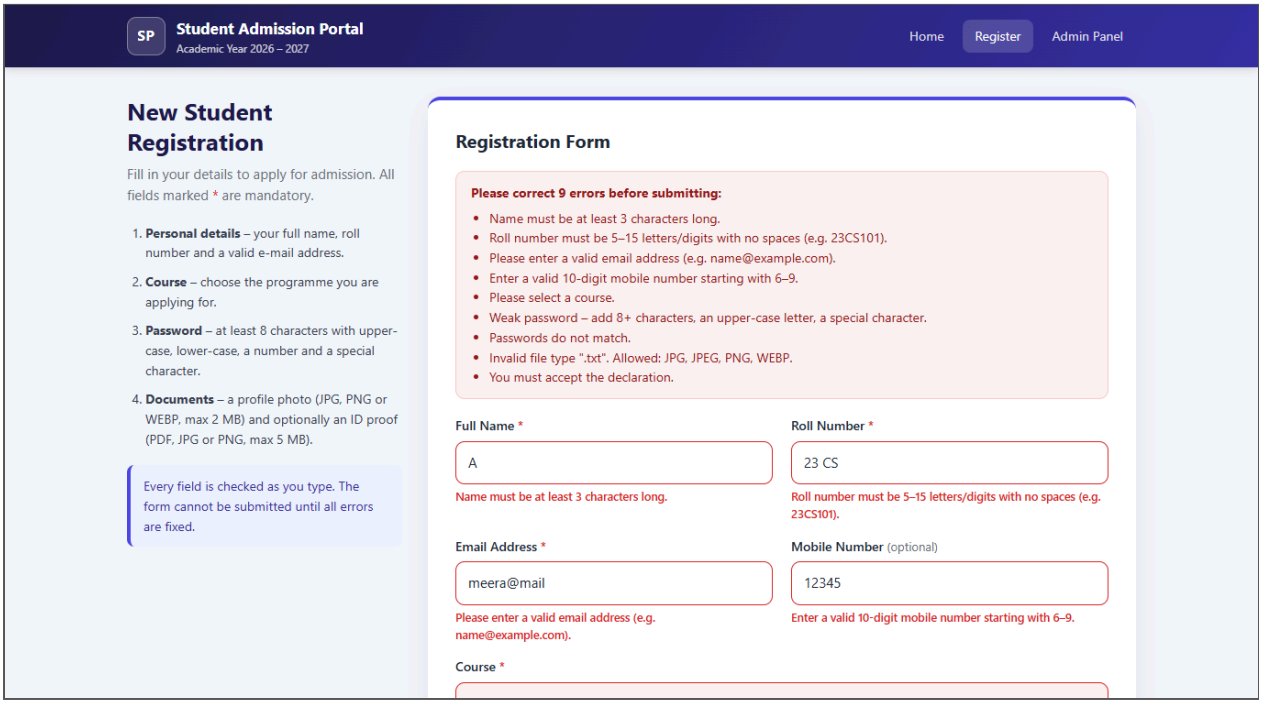
The form in `register.html` collects the student's name, roll number, email, mobile number, course, password, confirm password, profile photo and an optional ID proof. `style.css` gives a clean card layout with styled input fields, buttons and red error messages, and changes to a single column on mobile screens. `validation.js` checks every field before the form is submitted:

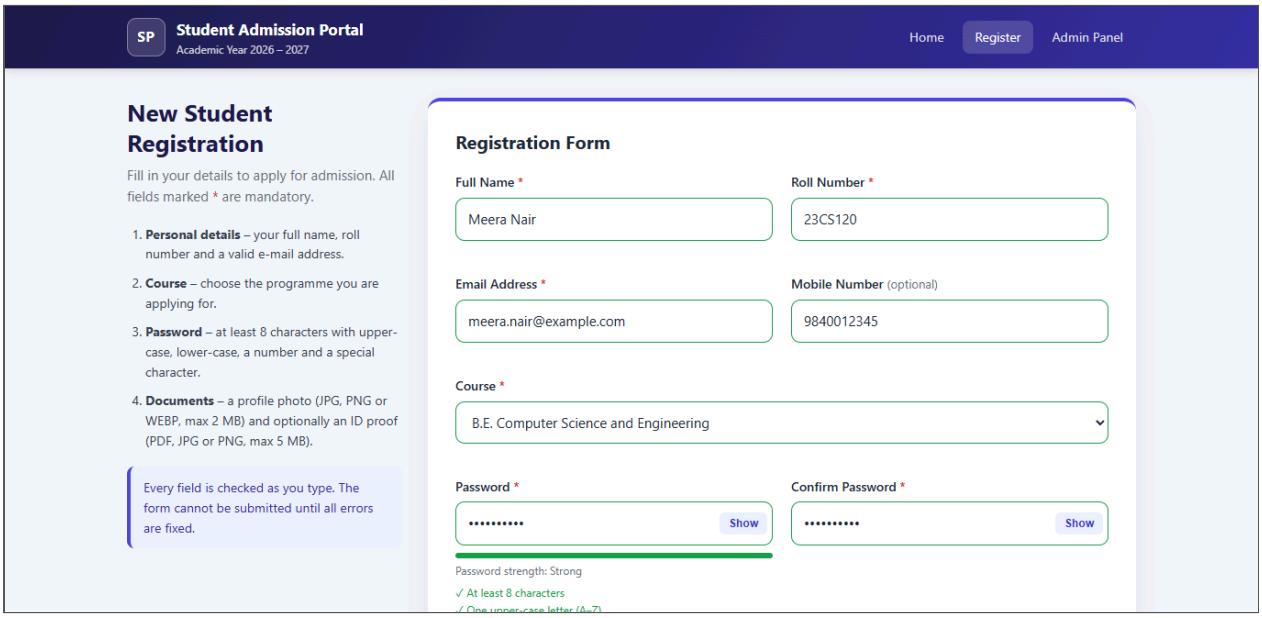
- **Required fields** – name, roll number, email, course, password and profile photo.
- **Email** – must be in the form *name@domain.com*.
- **Password strength** – at least 8 characters with an uppercase letter, a lowercase letter, a number and a special character, shown as a live strength meter.
- **Confirm password** – must match the password.
- **Files** – photo must be JPG/PNG/WEBP up to 2 MB; ID proof must be PDF/JPG/PNG up to 5 MB.

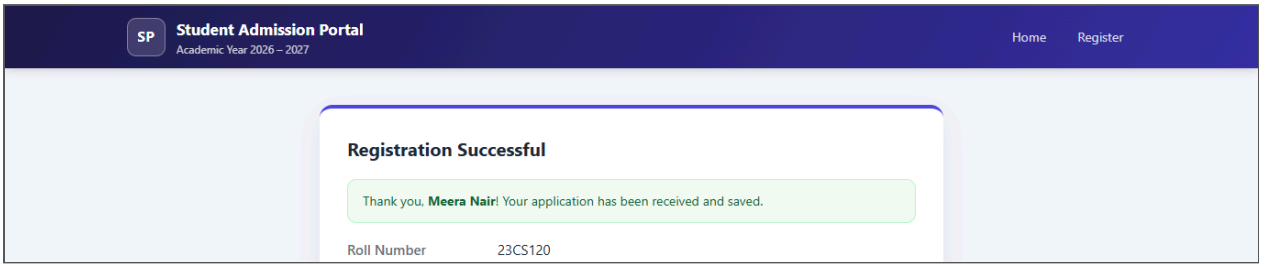
Code Explanation

HTML contains the labelled form fields with an error message area under each one. JavaScript has one validation function per field; when Submit is clicked every field is checked and `event.preventDefault()` stops the form if anything is wrong. Comments in the source files explain the important parts.

Output







Result

The registration form was created with a responsive design, and JavaScript validation checks the required fields, email format, password strength and uploaded files before submission.

QUESTION 2 – PHP & MySQL Admin Panel (16 Marks)

Aim

To manage student details and their profile images and documents using PHP and MySQL.

Implementation

The admin logs in to the panel and manages the student records stored in MySQL. All database queries use PDO prepared statements.

CRUD Operations

- **Create** – `add_student.php` adds a student with an optional photo and document.
- **Read** – `dashboard.php` lists all students with search; `view_student.php` shows one student's details and files.
- **Update** – `edit_student.php` edits the student details.
- **Delete** – `delete_student.php` asks for confirmation and removes the student and their files.

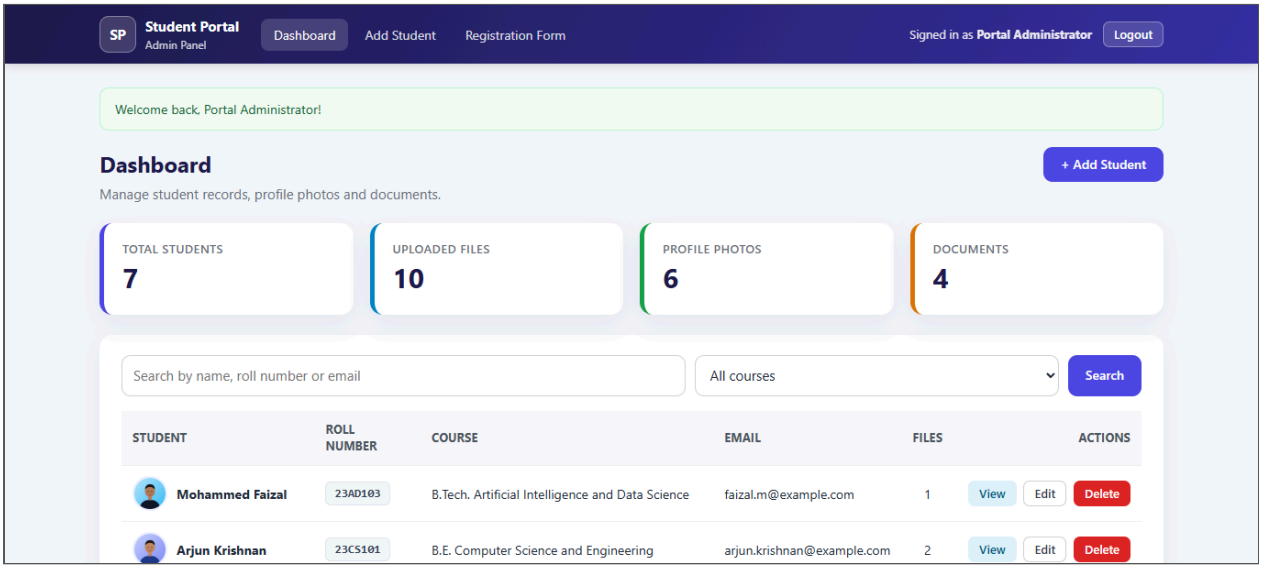
File Upload

Profile images and documents (ID proof, certificates) can be uploaded. Each file is checked for upload errors, size (2 MB for photos, 5 MB for documents), extension and real file type. It is saved with a random unique name, so an existing file is never overwritten. The file path, file type, size and upload date are stored in the `student_files` table, linked to the student, and are displayed when the student is viewed.

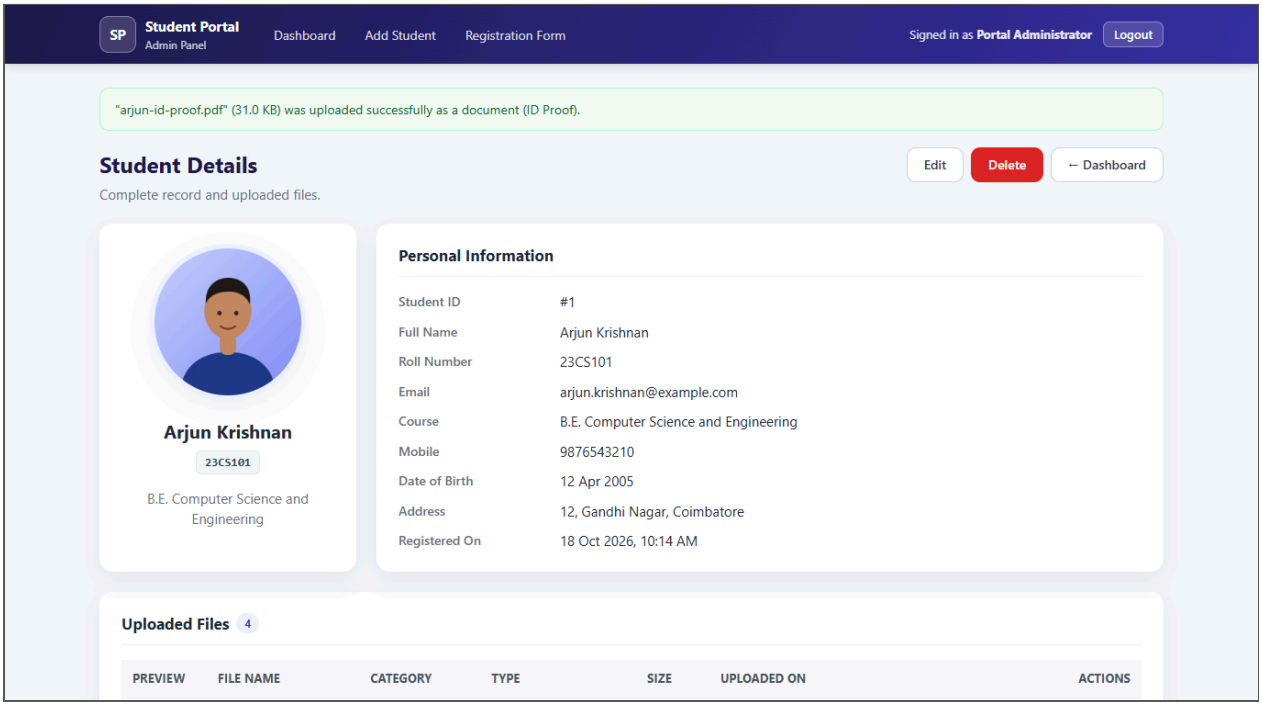
Code Explanation

`config/database.php` connects to MySQL using PDO, the CRUD pages run prepared SQL statements, `includes/upload_handler.php` validates and stores the uploaded files, and `view_file.php` shows a stored file only to a logged-in admin. Inline comments are included in the source files.

Output







Result

The PHP and MySQL admin panel was created with Add, View, Edit and Delete operations and secure file uploads, and the file details are stored in MySQL and displayed for each student.

QUESTION 3 – Servlet, JSP & JDBC (16 Marks)

Aim

To display student profiles and their uploaded files from MySQL using Servlets, JSP and JDBC with login protection.

Implementation

The Servlet project connects to the same MySQL database through JDBC (`DBConnection`, `StudentDAO`, `FileDAO`) and runs on Apache Tomcat.

JDBC / Servlet / JSP

`StudentListServlet` fetches all students and sends them to `students.jsp`, which loops through them with `<c:forEach>`. `StudentProfileServlet` fetches one student and all their files for `student-profile.jsp`, which uses `<c:choose>` to show an image preview or a PDF icon for each file.

FileDownloadServlet

`FileDownloadServlet` streams images and documents from the upload folder. The `<img>` tags in the JSP pages point to `file?id=…`, so the real file path is never shown in the browser.

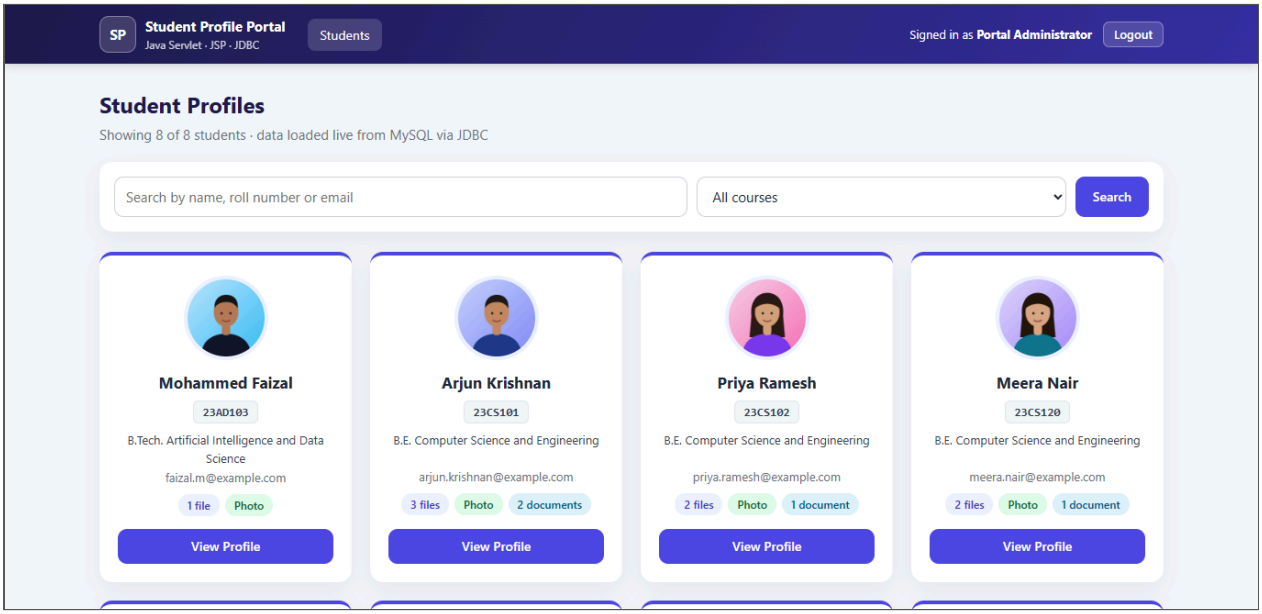
Session Handling

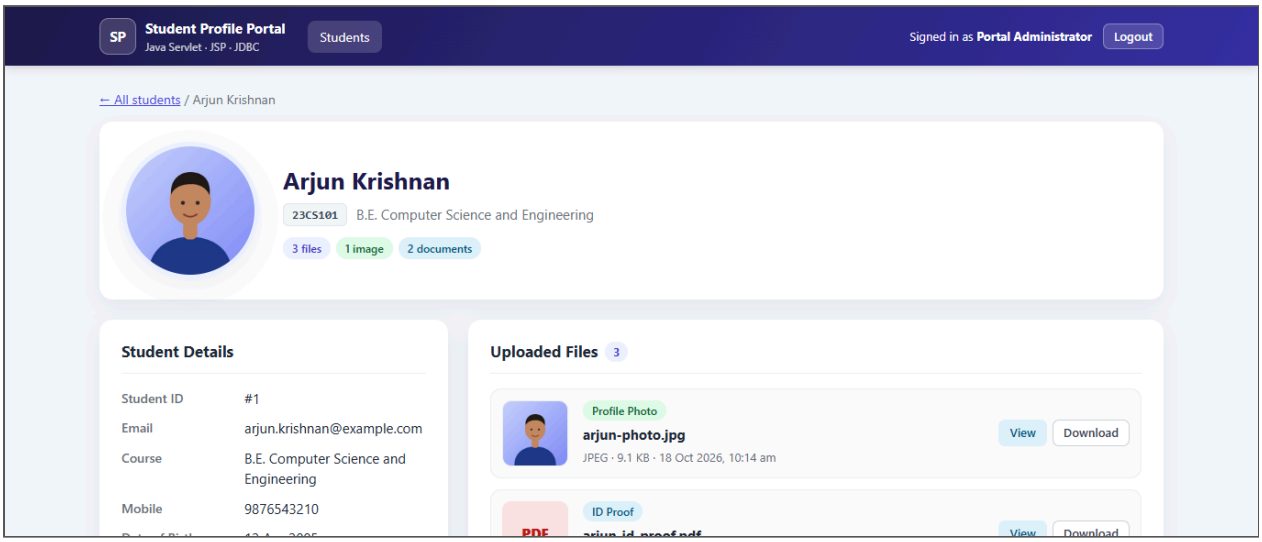
`LoginServlet` checks the username and hashed password and creates an `HttpSession`. Every protected page checks the session and redirects to the login page if the user is not logged in, and `LogoutServlet` ends the session.

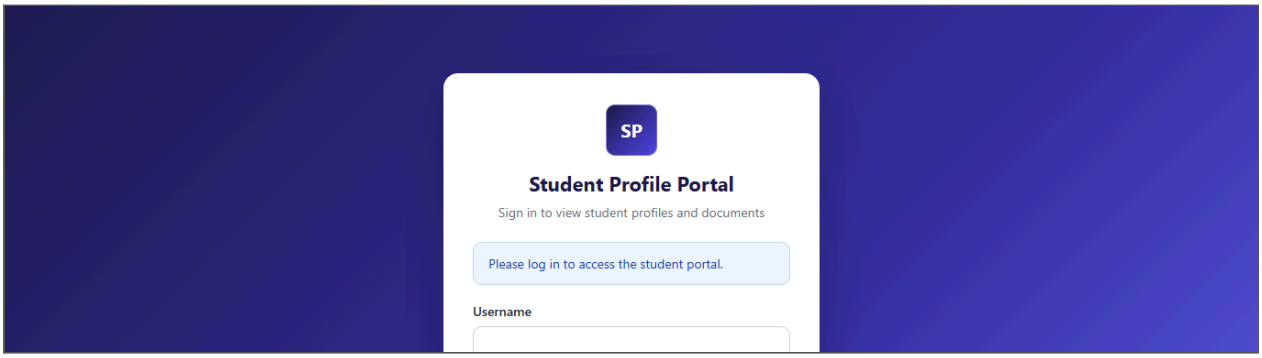
Code Explanation

The Java and JSP files contain the JDBC, Servlet, file-streaming and session logic. Inline comments are included in the source files.

Output







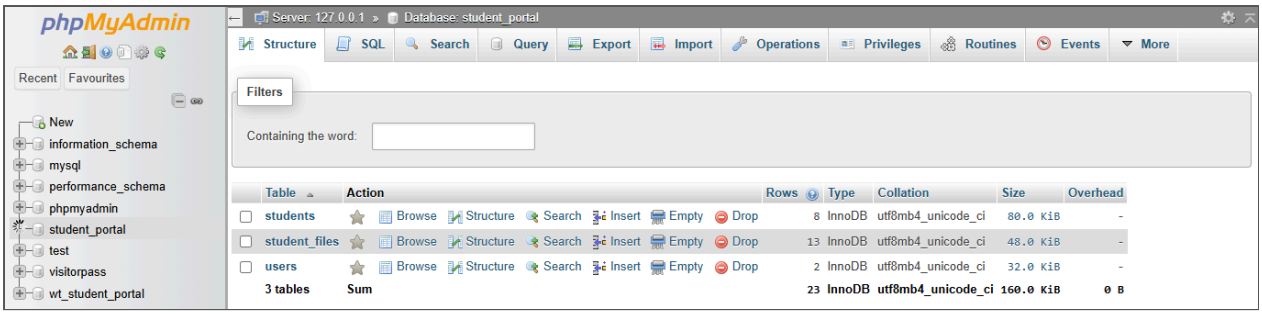
Result

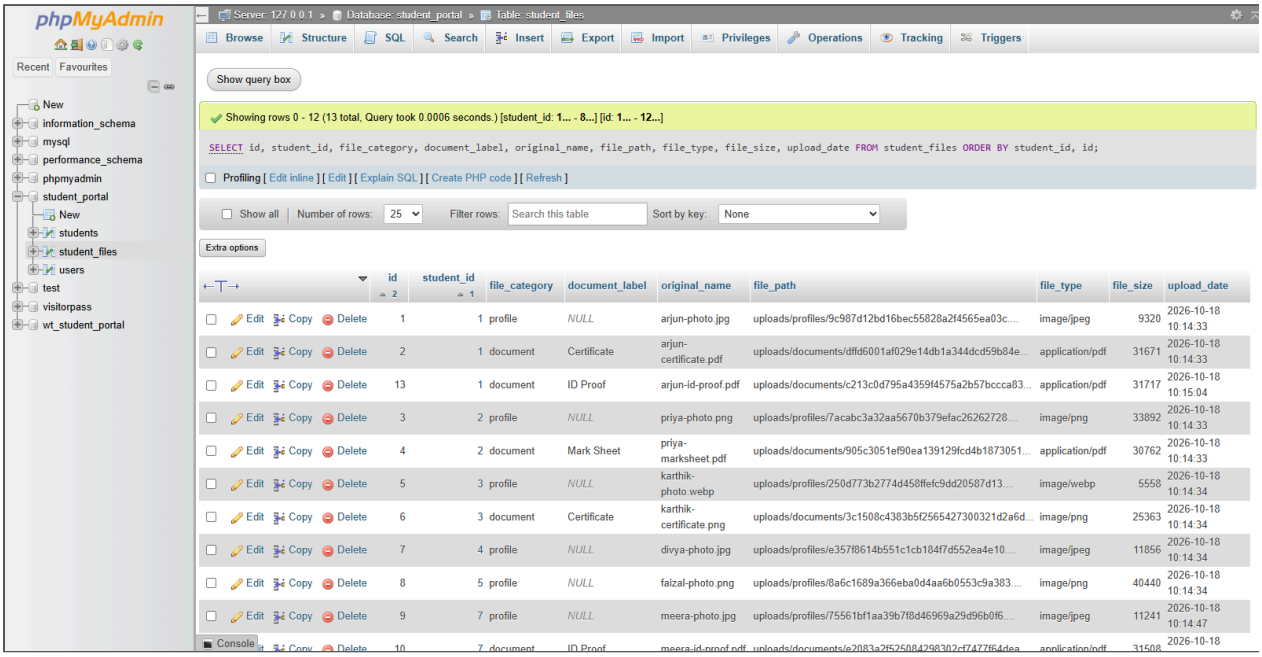
The Servlet and JSP portal displays student details and their uploaded files from MySQL using JDBC, streams images and documents through a Servlet, and allows only logged-in users to access it.

Database Schema

The database used is `student_portal`. The schema is available in `database/schema.sql`.

- **students** – Stores student details (name, roll number, email, course and others).
- **student_files** – Stores uploaded file details (student ID, file path, file type, size and upload date).
- **users** – Stores login details with hashed passwords.





Conclusion

Through this assignment, I learned to create a responsive registration form and validate it with JavaScript. I used PHP and MySQL to manage student records with secure file uploads, and Servlets, JSP and JDBC to display the same data with login sessions. This helped me understand how the front end, back end and database work together in one application.

GitHub Repository

https://github.com/Mithu2706/WT_Assignment – contains the complete source code, database schema and screenshots.